

FIRST PROGRESS REPORT – March 12, 2026

Project: CFPB Complaint Analysis (*Data Vintage - Feb 2024 to Feb 2026*)

Aim: Model customer complaint data using NLP and HMM (scope evolved from churn prediction to complaint triage and escalation modeling due to data constraints)

Tasks Accomplished:

1. Data cleaning, EDA and complaint text preprocessing.
2. Experimenting with various NLP techniques – Bag of words, n-grams, VADER, finBERT, RoBERT-a models.
3. Computing perplexity, determining optimum number of topics and validation of unigram vs bigram for topic modeling using LDA.
4. Finalized on these fronts – VADER features for sentiment analysis, 50 topics modeled using bigrams and LDA creating a final dataset for NLP workflow.

Current Output: ~1.4M complaints with validated NLP features

Immediate Next Steps:

1. As I am done with Phase-1 (NLP signals), I will validate these features using a baseline and a target variable.
2. I am planning to use the label 'Company response to consumer' due to absence of any clear churn-based binary classification flag. This has 5 classes and is readily available in the dataset.
3. Evaluate whether to use all 5 classes or consolidate into 3 meaningful tiers (Explanation Only / Service Recovery / Compensation) representing service complexity levels.
4. Finalizing target definition and model optimization through hyperparameter tuning (grid search).

Challenges and Adjustments:

1. No ground truth churn labels

Original Proposal : Predict customer churn from complaint text

Issue : CFPB dataset lacks customer identifiers and churn outcomes

Adjustment : Reframing as multi-class complaint triage (predicting service complexity tier) using comprehensive NLP features including topic modeling, sentiment analysis, and linguistic markers.

2. No customer-level sequences for traditional HMM

Original Proposal : Use HMM for defining different customer satisfaction stages (for churn prediction)

Issue : CFPB has complaint ids but no customer ids, so cannot observe customer sequences directly

Adjustment : Implementing simulation-based approach where I will hypothesize links between existing complaints using heuristics like same company, topic, timing, etc. and modeling complaint escalation pathways using HMM instead of transitions in customer satisfaction states.

Overall plan for project completion:

The most crucial step of the project now is to create a valid target definition and use a multi-class supervised learning approach to train, validate and select best model based on features derived till now. Further, prioritizing model optimization – tuning and temporal validation (OOS and OOT validation). Since the dataset lacks customer identifiers, I will create synthetic links between existing complaints to simulate state transitions and then train HMM to model complaint escalation pathways and enable scenario analysis for the previously defined target classes. Since the HMM phase involves simulation, synthetic link creation and training the HMM, it requires meticulous design and more attention. As it is the last phase of my proposed project, I will try to accomplish it within next 4 weeks, but it is also vital to document and create the final report in a detailed manner. Expected end goal for me is to deliver an intelligent complaint triage system combining supervised prediction and simulation-based escalation modeling. This can assist in intervention and optimization strategies for businesses to resolve and prioritize the complaints to avoid escalation and monetary loss and maintain reputation. However, if HMM implementation for full dataset proves too complex in a constrained timeframe, I intend to complete supervised classification with temporal validation fully and then try HMM for a smaller subset as a learning exercise and document HMM results for that sample.